Initial Thesis Plan: How does Cyclical Change Impact Functional Diversity, Ecosystem Response and Resilience across Ecoregions?

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Abstract

- 300 words
- set scene and introduce topic
- explain research, working hypothesis and main knowledge gap
- summarize work programme (methods) without extensive details

1 Introduction

1.1 Background

- 3000 words max
- what do we know?, set work in wider global context
- critical analysis of exisiting literature
- what do we not know? (knowledge gap and novelty of your work)
- allow reader to understand scope of project
- references and 2-3 figures/schematics/diagrames
- divided into subsections
- why is it good to study functional diversity? what do people study it for? e.g. ecosystem services, more informative than species richness
- key problem: Biodiversity is not static (changes with seasons (migration, hibernation) even daily as nocturnal species replace the diurnal)), we lose important information about how these ecosystem services vary across time and space, species interactions, niche partitioning (temporal as well niche) etc if we fail to characterise functional turnover. Further we don't know at what temporal granularity these ecosystem services shifts might/need to be captured.
- A lot of work has been done looking shift in biodiversity across space e.g.latitudinal shifts, and along environmental gradients but not so much across the annual cycle, particularly outside of coarse seasonal change (i.e. winter vs summer).
- To address this knowledge gap, this study will combine functional diversity with cyclical change to assess how seasonal and daily turnover effects ecosystem functioning and resilience to global change
- We will expect to see (hypothesis background below)

2 Aims and Objectives

- 300 words

2.1 Aims

- 1-3 aims, that address knowledge gaps previously stated. one sentence per hypothesis

1. **chapter 1: Characterising how migration affects functional diversity**
2. **chapter 2: Characterising how day to night shifts affect functional diversity**
3. **chapter 3: Understanding how cyclical change affects non-avian taxa**
4. **chapter 4: Understanding how climate change impacts cyclical change and ecosystem functioning**

2.2 Hypotheses

Hypothesis 1: In birds, migration throughout the annual cycle is important for ecosystem functioning. However, migrant functional contributions will vary with season, dietary/foraging niche, duration of stay, ecoregion and latitude

- on the one hand may expect migrants to be contributing specialist functions since they tend to specialise on particualar resources which results in the evolution of migration as a strategy in the first place, or may have particualr climatic niches which neccessitate migration across seasons/wet/dry periods
- migration also requires unique traits e.g. HWI/bodymass (lots of evidence for this; but see Williams and Curley, 2025)
- also migrants need to be somewhat divergent in their traits - limiting similarity hypothesis, to avoid competition with residents/other migrants?
- may also expect migrants to be more spread out, as they are likely less generalist and plastic (residents may switch their niche across seasons), occupying more extreme trait combinations
- However, there is also likely to be a lot of overlap with resident communities as both migrants and residents are captilising on similar resource peaks that are more abundant at particular times of year. e.g. an african bird migrating north for seeds will not suddenly eat fish, it will probably eat the abundant seeds that the residents are likely also captilising on. Monopolisation of these resources is a different quesiotn e.g. Williams and Curley, 2025
- However, this does not necessarily mean that migrant redundancy and complementarity are mutually exclusive e.g. In north mediterrannean, flamingoes come to eat abundant shrimp in the salt farms, a resource not exploited by the resident community (or other migrants for that matter)

- This functional overlap may also be the case across seasons- evidence for biotic homogenisation of migrants across seasons, probably due to shifting migrant species distributions and changing resource peaks (i.e. phenological mismatch).

- Additionally, more transient migrants are likely to be more unique as they may be migrating long distances resulting in the need for short stop-overs (calculate migratory distance as well to confirm this). Thus, these short-stay species are likely have diet/foraging types included in their niche that are not present in their stop-over region, and therefore they may have unique trait combinations that are distinct from resident communities/long-stay migrants that are not utilising these "foreign resources" in other locations/parts of the year

- however, this is unlikely to be significant for ecosystem functioning because a species present in a month is unlikely to make a signifcant functional contribution

- functional contribution of migrants will differ across seasons, dietary/foraging types, and ecoregions

- i.e. previously granivores/frugivores found to dominate in winter, whilst insectivores are more prevalent in summer (Keyser et al., 2024; Bellve et al., 2025)

- additionally, nestedness vs overlap with resident communities tends to differ across habitat types (Oliveira et al., 2023)

- there is also a latitudinal effect of seasonality, i.e. seasonal and resource variation is greater at northern latitudes and migration is more dominant in the northern hemisphere,

- thus migrants are likely to contribute more to functional diversity at higher latitudes, potentially introducing more specialist functions.

- On the other hand competition may be greater in the tropics where species richness is greatest, therefore migrants and residents may be able to better partition resources in seasonal environments (higher latitudes) than non-seasonal environments.

- Thus, migrants may be more specialist in the tropics due to the limiting similarity hypothesis (although greater producitivty may also facilitate niche packing of migrants and residents)

Hypothesis 2: In birds, ecosystem functioning will shift across the daily cycle and birds unique to the nocturnal or diurnal niche may represent unique functions.

- species that are purely nocturnal/diurnal likely contribute unique and/or specialist traits e.g. beak shape for nectar (lots of diurnal species) vs rodents (lots of nocturnal species) will differ and species will be using different senses e.g. heat vs colour vision which will impact their trait shape/size and this will impact ecosystem functioning, niche space and species interactions.

- species existing in both night and day likely feed on non-colour reliant resources. or may be more plastic in their niche (generalists)

Hypothesis 3: In mammals/other taxa, changes across the annual cycle (i.e. migration) will likely play as less significant role than in birds (although hibernation in mammals may be important), however daily functional shifts may be more important.

- migration is much more widespread in birds (40% percent of birds migrate whilst only 20% mammals migrate- ungulates and marine mammals are really the only migrating mammals and in amphibians, migration is will be more specific to breeding season and probably much more relevant for local aquatic functional diveristy (ponds))
- but daily changes are likely more important as activity period seems to be more important part of ecology in other taxa, with more nocturnal mammals? (see Cox and King et al papers)

Hypothesis 4: Climate-induced shifts in species' distributions will disproportionately effect migrants and species specific to one acitvity period as they may be more specialist and less adaptable under climate change, impacting ecosystem functioning

- migrants/activity period specialists likely have more specific climate-niches than residents, driving their migratory/acitvity period life-history in the first place. Thus, they may be more sensitive than residents or acitvity period generalists to climate change.
- if they are found to contribute unique functions (in above chapters), this may have real consequences for ecosystems.
- Additionally, the migrant/ acitivty period species gained in ecoregions are likely to be the more plastic niche switchers/ generalists which will probably be more functionally similar to the existing resident community/species existing across both nocturnal and diurnal niches
- More direct forms of disruption will likely also effect cyclical species and therefore their functional contributions e.g. wind farms, dams, trainlines, artificial lights

2.3 Research Objectives

- 3-6 objectives, one sentence per objective., Main research goals you need to address your questions/hypotheses
- research goals you need to address your hypotheses-

1. Quantify migrant contribution to avian functional trait space (i.e. niche restructuring) across the annual cycle at increasing temporal granularity
2. Quantify nocturnal/diurnal species' contribution to the avian functional trait space across the 24hour cycle
3. Quantify migrant/hibernator contribution to the mammalian functional trait space across the annual cycle

4. Quantify nocturnal/diurnal species' contribution to the mammalian functional trait space across the 24hour cycle
5. Assess species' distributional shifts under climate change and how this might impact functional diversity given the loss/gain of cyclical species

3 Initial research plan

- 800 words
- Outline of research intent for chapter 1 (8-15 months of research or maybe less)
- state which hypothesis you are addressing
- describe analysis/modelling approach and methods
- how you intend to interpret the results/ draw conclusions
- diagrams/schematics useful with legends
- addressing hypothesis 1
- PCAS/PCOAs
- disprity metrics
- models

4 References